Lucid dreaming

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Glossary

Eye signaling technique An objective method for verifying lucid dreams through distinct volitional eye movement patterns recorded on an electrooculogram (EOG). The technique can also be used to signal the onset and offset of experimental tasks completed in the dream.

Galantamine An acetylcholinesterase inhibitor (AChEI) that increases the likelihood of lucid dreaming when taken late in the sleep cycle with an appropriate mental set.

Lucid dream A dream in which one knows that one is dreaming during ongoing sleep.

MILD Mnemonic Induction of Lucid Dreaming is a technique to induce lucid dreams using prospective memory.

Phasic activity A transient increase in physiological activation, including increased REM density.

Rapid eye movement (REM) sleep A stage of sleep characterized by an activated EEG showing mixed frequency desynchronized waveforms, submental EMG atonia, and complex combinations of "rapid" eye movements.

REM density The frequency of eye movements during REM sleep per unit time. REM density tends to be higher both during and just prior to the onset of lucid dreams.

WBTB Wake-Back-To-Bed is a technique to induce lucid dreams in which one awakens during the night, typically after 4–6 hours of sleep, remains awake for approximately 30 minutes and then returns to sleep.

Methodology

A new methodology in sleep and dream science

Lucid dreaming occurs when one recognizes that one is dreaming during ongoing sleep. Once lucid, an individual is typically able to exert volitional control over the dream, which offers opportunities for recreation, creativity, problem-solving, and practicing skills

 https://doi.org/10.1016/B978-0-12-822963-7.00035-9

for waking life (LaBerge and Rheingold, 1990). Lucid dreaming is both an emerging area of scientific research, as well as a promising method for the study of dreaming, sleep and, more generally, human consciousness. Foundational to the scientific study of lucid dreaming was the discovery that lucid dreams can be objectively verified on a polysomnogram (PSG) through volitional eye movements performed during REM sleep dreaming. Using this technique, research on lucid dreaming has yielded unique insights into the relationship between physiology and psychology during sleep. Preliminary data suggest that lucid dreaming may also be useful in clinical applications, particularly for patients suffering from nightmares.

Eye signaling technique

The first physiological studies of lucid dreaming were conducted in the late 1970s and early 1980s. These studies built on prior research demonstrating that shifts in the direction of gaze within a dream can be accompanied by corresponding movements of the sleeper's eyes (Dement and Wolpert, 1958). This insight led researchers to test whether lucid dreaming could be scientifically verified through recording pre-agreed upon eye movements during lucid REM sleep. Participants who reported frequent lucid dreaming slept in a sleep laboratory and made volitional eye movement signals if they had a lucid dream by rapidly looking left and right or up and down within the dream. This research confirmed that reports of lucid dreams could be objectively verified by the presence of distinct volitional eye movement patterns recorded by electrooculogram (EOG) during PSG-verified sleep. A left-right-left-right-center (LRLRc) signal, illustrated in **Fig. 1**, is now the gold standard and is used in laboratories all over the world. As can be seen in the figure, the eye signal is readily discernible in the horizontal EOG, which exhibits a distinctive shape containing consecutive full-scale eye movements that have larger amplitude compared to typical REMs. Lucid dreams are validated through the convergence between reports obtained after awakening of making the eye movement signals during a lucid dream accompanied by the objective eye movement signals recorded in the EOG with concurrent polysomnographic evidence of REM sleep.

Psychophysiology

Physiology of lucid dreaming

Using the eye signaling technique described in the previous section, initial experiments discovered that lucid dreaming occurs not only during clinically defined sleep, but during the REM sleep stage (Hearne, 1978; LaBerge et al., 1981). While there have been several reports of lucid dreams during NREM sleep, research indicates that lucid dreaming is predominately a REM sleep phenomenon. Further research documented that lucid dreams tend to occur during the most activated stage of REM sleep, with heightened phasic activity, including increased REM density. Markers of autonomic nervous system arousal (e.g., heart rate, respiration rate, and skin potential) also tend to be elevated during lucid REM sleep compared to baseline REM sleep (LaBerge et al., 1986). Additionally, lucid dreams more often occur in REM sleep periods later in the night, which is also associated with increased phasic activity and increased REM density. In addition to physiological markers of phasic activity, lucid REM sleep is associated with h-reflex suppression, a spinal reflex that is reliably suppressed during the REM sleep stage (Brylowski et al., 1989). This finding provides strong support for the conclusion that lucid dreaming occurs during REM sleep, as opposed to a state that is intermediate between REM sleep and wakefulness. Together these results indicate that lucid dreams occur in activated periods of REM sleep.

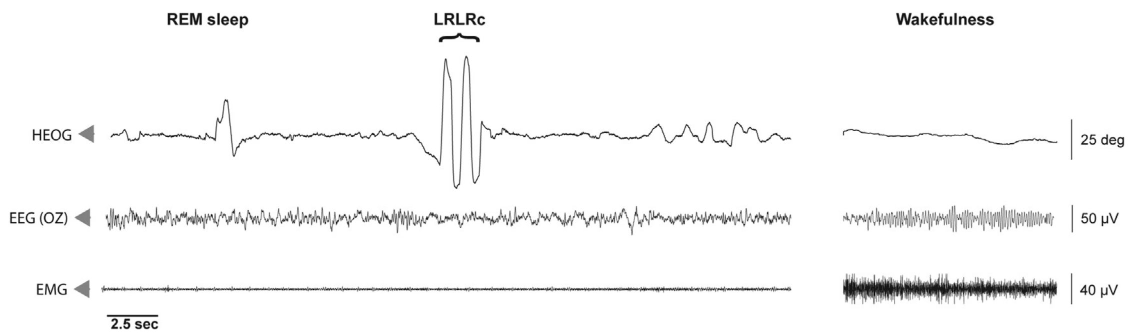

Fig. 1 Left-right-left-right-center (LRLRc) eye movement signaling performed during PSG-verified lucid REM sleep. Participants signal when they become lucid in the dream by quickly looking to the left (as if looking at their ear) and then all the way to the right two times consecutively without pausing. The LRLRc signal appears on the HEOG as consecutive full-scale high-amplitude eye movements. Note that the electroencephalogram (EEG) contains theta rhythm (~5 Hz), the OZ channel lacks alpha, and the electromyogram (EMG) has minimal amplitude due to muscle atonia, which is characteristic of REM sleep (left) as compared to wakefulness (right).

Electrophysiology of lucid dreaming

As discussed above, lucid dreaming is associated with multiple markers of brain activation and physiological arousal. However, whether lucid REM sleep is associated with activation of specific brain regions or changes in specific frequencies of neural oscillations remains an open question. Several EEG studies have attempted to address this question, although all of them have interpretive issues and most suffer from low statistical power. The findings from these studies have been mixed and conflicting.

Although early findings suggested increased alpha band power (8–13 Hz) during lucid dreaming (Ogilvie et al., 1978), subsequent attempts to replicate this finding were unsuccessful. Thus, current evidence does not support a reliable association between alpha band power and lucid dreaming (LaBerge, 1988). Another study found increased power in the low beta frequency range (13–19 Hz) in parietal electrodes for lucid compared to non-lucid REM sleep (Holzinger et al., 2006). However, limited spatial resolution and technical issues suggest the need for replication of these findings. Another study reported increased gamma band (40 Hz) power in frontolateral scalp electrodes during lucid compared to non-lucid REM sleep (Voss et al., 2009). However, the small sample size ($N = 3$) and the potential confounding influence of ocular myogenic artifacts limit interpretation of the results. Another study in narcoleptic patients found that EEG power in the delta band was reduced during lucid REM sleep at frontal and central electrodes (Dodet et al., 2014). However, only 6 channels were measured, and it remains unclear whether the results will generalize to a nonclinical population. Thus, together these four studies present conflicting findings regarding the localized EEG correlates of lucid dreaming. The discrepancies in EEG findings are likely due to several causes, including the limited electrode montages and low statistical power. Research with larger sample sizes and higher spatial resolution EEG will be necessary to shed light on these conflicting findings.

Neuroimaging of lucid dreaming

There has been only one fMRI study of lucid REM sleep, and it is a case study (Dresler et al., 2012). Two separate signal-verified lucid dreams were recorded from the subject during EEG-verified REM sleep inside the MRI scanner. Lucid REM sleep showed increased fMRI BOLD signal in several cortical regions, such as the anterior prefrontal cortex, medial and lateral parietal cortex, inferior/middle temporal gyri and occipital cortex. Interestingly, these frontoparietal regions are regions that consistently show reduced cerebral blood flow during REM sleep as compared to wakefulness (Maquet et al., 1996). However, the activations may be partially confounded by the fact that the subject was performing an experimental task during lucid REM sleep. A group-level fMRI study using a no-task paradigm is therefore needed to further evaluate regional BOLD signal changes associated with lucid REM sleep.

Validation/research examples

Eye signaling as an experimental methodology

The eye signaling technique offers a way of objectively verifying lucid dreaming as well as contrasting the physiology of lucid REM sleep to baseline REM sleep. It also provides a powerful methodology for the science of sleep and dreaming. Specifically, lucid dreamers can not only signal to indicate that they are aware that they are dreaming, but they can also make the eye movement signals to time-stamp the start and end of experimental tasks performed during lucid dreams. By providing objective temporal markers recorded on an ongoing PSG recording, this technique offers a new method for studying the psychophysiology of REM sleep, allowing for investigations into the neural correlates of dreamed behaviors. Lucid dreaming thus provides a way to identify precise psychophysiological correlations between psychological content during sleep and physiological measures that opens up new possibilities for research. In this section, we briefly highlight several recent examples of how this methodology can be used to answer questions in sleep research and cognitive neuroscience.

Sensorimotor cortex activation during dreamed movement in REM sleep

Employing the eye signaling methodology, researchers have tested whether clenching one's hand in a dream activates the sensorimotor cortex (Dresler et al., 2011). During a lucid dream, participants performed 10 left hand clenches, then LRLRc eye signaling. They repeated this pattern for the right hand and continued the sequence as long as possible before awakening. While awake, participants performed the hand-clenching task as well as imagined the task for comparison. A localized increase in the BOLD signal was seen in the contralateral sensorimotor cortex in response to both the left and right hand clenches during lucid dreaming. BOLD signal fluctuations during dreaming were only approximately 50% the size of waking fluctuations, potentially as a result of the lack of sensory feedback due to REM sleep atonia. Overall, the data show that patterns of brain activation observed during dreamed movements closely overlap with motor execution during wakefulness, despite being weaker and more localized. While the study was limited by having only two case reports, it provides a proof-of-concept for utilizing lucid dreaming to examine the neural activity of specific dream content using fMRI.

Voluntary control of central breathing apneas during REM sleep

Lucid dreaming has also been used to test whether irregular breathing during REM sleep has a cortical origin (Oudiette et al., 2018). Polysomnographic and respiratory data were collected from 21 patients with narcolepsy who reported frequent lucid dreams.

Participants performed LRLRc eye signaling and then modified their dream scenario to include behaviors that require cortical control of respiration (e.g., diving under water). Participants also performed the tasks during waking imagination for comparison. A total of 32 lucid REM sleep naps were included for analysis, although only 16 showed the expected respiratory behavior. These data suggest that the cortico-pontine drive that regulates the cessation of breathing is preserved at least part of the time during the REM sleep state.

Smooth pursuit eye movements during visual tracking in REM sleep

Lucid dreaming has been used as a method to study smooth pursuit eye movements during REM sleep (LaBerge et al., 2018a). Participants with high dream recall and lucid dream frequency underwent sleep laboratory recordings in which they performed LRLRc eye signals upon becoming lucid in the dream, and then tracked the movement of the tip of their (dreamed) thumb while either tracing a circle or moving it back and forth horizontally along the horizontal meridian. Participants performed the tracking tasks while in lucid REM sleep, while awake with eyes open, and while tracking an imagined version of the movement while awake with eyes closed. Slow tracking of visual motion (both circles and lines) during lucid REM sleep did not differ from pursuit during waking tracking. Both conditions were characterized by high pursuit ratios and low saccade rates. Conversely, the imagination tracking condition was characterized by low pursuit and frequent catch-up saccades. These findings suggest that the visual imagery of REM sleep dreams is more similar to perception than to imagination and that shifts in eye movements during REM sleep mirror the direction of a subject's gaze in dreams. An important implication of these findings for the neuroscience of smooth pursuit is that neither a physical motion stimulus nor readout of retinal image motion is necessary to generate smooth pursuit eye movements.

Clinical applications

Nightmares

The most researched application of lucid dreaming is to treat nightmares. Lucid dreaming is included by the American Academy of Sleep Medicine as a therapy suggestion for nightmare disorder (Morgenthaler et al., 2018). Preliminary controlled trials suggest that lucid dreaming therapy can be effective in reducing nightmare frequency. For example, one study of 32 patients who suffer from frequent nightmares found a slight advantage of lucid dreaming as add-on to Gestalt therapy compared to the latter alone (Holzinger et al., 2015). The highest promise of this approach is in a therapeutic context where a patient can, once becoming aware that it is a dream, actively work with the psychological content of the nightmare. Neurocognitive models of nightmare generation suggest a hyper-responsivity of the amygdala coupled with a failure of prefrontal regions to dampen the amygdala. Studying the influence of lucidity on nightmare resolution could therefore provide a model by which to study top-down regulation of amygdala activation. Controlled experiments with larger sample sizes are needed to further evaluate the potential usefulness of lucid dreaming for the treatment of recurring nightmares and related disorders, such as nightmares in post-traumatic stress disorder (PTSD).

Narcolepsy

Patients with narcolepsy report that becoming lucid in their dreams can provide psychological relief (Rak et al., 2015). Generally, narcolepsy patients experience longer, more complex, and more vivid dreams and more frequent nightmares than healthy subjects. Moreover, patients with narcolepsy report higher lucid dreaming frequency, though the mechanisms for this are not well understood at this time. Nightmares can often lead to insight into the current dream state; however, it is unknown whether an increased nightmare frequency causes an increased lucid dreaming frequency in narcolepsy, or if the strongly increased frequency of REM sleep in narcolepsy affects nightmares and lucid dreaming independently. A related possibility is that sleep onset REM (SOREM) episodes in narcoleptic patients may prompt lucid dreams, which is in line with the effects of the early morning "Wake-Back-to-Bed" procedure, described below, in healthy subjects.

Induction procedures

Induction methods overview

A main target of research is to develop methods to make the lucid dream state more accessible. While most people rarely experience lucid dreams, there is substantial variation in lucid dream frequency, ranging, by current estimates, from never (approximately 40%–50% of the general population) to monthly (approximately 10%–20%) to a very small percentage of people who report lucid dreams weekly, nightly or multiple times per night (<1%). Given that lucid dreaming occurs very infrequency for most individuals, reliable techniques to induce lucid dreams are necessary for it to be effectively used in both clinical and scientific applications. Research suggests that lucid dreaming is a learnable skill that can be developed by training with various induction strategies. These include training in prospective memory techniques, which can be further aided by application of external sensory cues and/or interrupting sleep with periods of wakefulness. As discussed below, pharmacological interventions can also substantially increase the likelihood of lucid dreams. Importantly, sleep interruption, sensory cueing and pharmacological interventions are not alternatives to prospective memory, but rather are further aids to an appropriate mental set.

Prospective memory

One of the most effective means of inducing lucid dreams is to set an intention to remember that one is dreaming before falling asleep or during awakening, ideally with respect to specific dream content, thus engaging prospective memory. Many induction methods utilize prospective memory, often in combination with visualizations and/or repeated phrases. For example, the Mnemonic Induction of Lucid Dreams (MILD) technique employs prospective memory in four stages (LaBerge and Rheingold, 1990). First, the individual recalls a recent dream and mentally relives the experience of the dream. Next, they identify dream-like characteristics of dream, such as rapid scenery changes, lapses in memory and unrealistic events. Third, they practice recognizing these events as evidence that they are dreaming, leading them to realize that they are dreaming. Finally, they visualize acting in a lucid dream and rehearse any goals. The MILD technique is most effective when performed immediately before a dream.

Sleep interruption

Periods of extended wakefulness (30 minutes or more) during the night increase the likelihood of lucid dreams after returning to sleep (LaBerge et al., 1994). This technique is often referred to as Wake-Back-To-Bed (WBTB). Participants awaken approximately 4.5–6 hours into their sleep period either spontaneously after an awakening from REM sleep cycle, or, if that is not feasible, by setting an alarm. The aim is to awaken after approximately the end of the third or fourth REM sleep cycle. There may be several reasons why this technique is effective.

First, since most lucid dreams occur later in the night, WBTB primes participants to be more critical and aware before the period of sleep in which they are mostly likely to have a lucid dream. Relatedly, as the sleep cycle progresses, REM sleep periods become both longer and closer together, meaning that participants have a good chance of quickly entering the REM sleep dreaming state shortly after the period of sleep interruption. Finally, the period of awakening may result in increased brain activation in several brain areas, which could contribute to higher lucid dreaming frequency, though at present this idea remains speculative.

External sensory cues

Another approach to lucid dream induction introduces sensory stimuli to individuals in REM sleep. This approach utilizes the fact that some types of sensory stimuli can be incorporated into ongoing dreams from the sleeping environment. This can include sounds, lights, odors and tactile sensations. Several devices have been developed to take advantage of this phenomenon to induce lucid dreams, such as sleep masks that flash lights and play audio when the dreamer enters REM sleep. This type of sensory incorporation provides a reliable and consistent mnemonic cue that an individual can learn to recognize as a signal that they are currently dreaming using prospective memory (e.g., LaBerge et al., 1988).

Pharmacological interventions

As discussed above, lucid dreaming is associated with increased cortical activation, which reaches its peak during phasic REM sleep. Given the role of Acetylcholine (ACh) in REM sleep regulation, agents acting on the cholinergic system have the potential to influence cortical activation and increase the frequency of lucid dreams. LaBerge et al. (2018b) conducted a double blind, placebo-controlled study in a large group of participants ($N = 121$) with high dream recall and an interest in lucid dreaming. The study tested the effect of galantamine, a fast-acting AChEI, taken during a period of sleep interruption combined with the Mnemonic Induction of Lucid Dreams (MILD) technique. Galantamine significantly increased the frequency of lucid dreaming in a dose-related manner from approximately 14% in the placebo group to 27% with the half dose (4 mg) to 42% with the full dose (8 mg). Another double blind, placebo-controlled study also tested the effect of galantamine on lucid dreaming (Sparrow et al., 2018). 35 participants completed an eight-night study that tested the effect of 8 mg of galantamine paired with 40 min of sleep interruption. Galantamine significantly increased lucid dream frequency compared to several control conditions with similar percentages to LaBerge et al. (Baird et al., 2019). Galantamine is currently the most effective pharmacological intervention for lucid dream induction when used in combination with sleep interruption and prospective memory techniques.

Induction of lucid dreams with transcranial electrical brain stimulation

Two studies have attempted to induce lucid dreams through transcranial electrical stimulation of the frontal cortex during REM sleep. In a study that tested the effect of direct transcranial current stimulation (tDCS), researchers applied either tDCS or sham stimulation (counterbalanced across nights) over a frontolateral scalp region during REM sleep periods in 19 participants (Stumbrys et al., 2013). tDCS did not significantly increase the number of dreams rated by judges to have a clear indication of lucidity: seven dreams in total, three from sham stimulation and four from tDCS stimulation. Furthermore, only one lucid dream was confirmed by eye signaling, which was in the stimulation condition. Given a difference of only one lucid dream between stimulation and sham conditions, overall stimulation with tDCS as tested in this study does not appear to be a reliable method for inducing lucid dreams.

Another study reported a pronounced increase in lucid dreaming by applying transcranial alternating current stimulation (tACS) in the gamma frequency band over the frontal cortex (Voss et al., 2014). The study tested 27 participants for up to four nights in the sleep laboratory. Participants were then awakened and completed the Lucidity and Consciousness in Dreams (LuCiD) scale.

Importantly, lucid dreams were not assessed with eye signaling, self-report, or through statistical analysis of judges' ratings of dream reports. Rather, dreams were assumed to be lucid if subjects reported elevated ratings on either or both of the LuCiD scale factors insight and dissociation. Mean ratings in the insight subscale increased from approximately 0.1–0.2 in the sham stimulation to 0.5–0.6 in the 25 Hz or 40 Hz stimulation conditions, similar to the effects in the dissociation subscale, which was approximately 0.9–1.2. However, the scale anchors ranged from 0 (strongly disagree) to 5 (strongly agree), indicating that, on average, in both the 25 Hz and 40 Hz stimulation conditions, participants disagreed that their dreams had increased insight. Moreover, a recent study failed to replicate induction of lucid dreams with this technique (Blanchette-Carrière et al., 2020). Overall, the results of this study appear to be comparable to direct tDCS stimulation. In summary, a method for reliably inducing lucid dreams by electrical stimulation of the brain is yet to be identified.

Conclusion

Lucid dreaming is an emerging methodology in sleep and dream science which shows promise in both clinical and scientific contexts. The eye signaling technique offers a novel method for research on dreamed behaviors and content. Lucid dreaming occurs during REM sleep and is associated with increased brain activation. It is currently unknown whether lucid dreaming is associated with activation of specific brain regions or changes in specific frequencies of neural oscillations. Future research with larger sample sizes and higher-resolution spatial sampling are necessary to elucidate the electrophysiological correlates of lucid dreaming. Clinically, lucid dreaming offers a new avenue to treat nightmare disorder and may have applications to other sleep-related disorders. Research on induction strategies aims to make lucid dreaming more accessible for scientific, clinical and personal applications. A combined induction strategy using prospective memory training, sleep interruption and acetylcholinesterase inhibition is currently the most effective method for inducing lucid dreams. Altogether, lucid dreaming provides a useful methodology for exploring the relationships between sleep physiology, psychology and human consciousness.

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